Group 2
MIT805: Big Data Project (2026) - Part 1

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Project Title: Stock Market Data Exploratory Data Analysis

1. Dataset Provenance and Description

1.1. Dataset Source, Dataset License and terms of Use

The dataset was published on Hugging Face (stock-market-data-warehouse) and the dataset metadata exist on the Github repository (stock-data repo) by **brandonyeequon.** The dataset is licensed under the MIT Licence which is open source and the author has specified that the dataset can be used for academic research. The original source of the dataset is Free Stock APIs | Alpha Vantage which is a NASDAQ-licensed data provider.

1.2. Dataset Description:

Dataset Description: The dataset is about 15-minute interval OHLCV (Open, High, Low, Close, Volume) of all major US exchanges made up of 6564 stock tickers across 12 sectors which include Consumer Discretionary, Consumer Staples, Energy, Financials, Basic Materials, Healthcare, Industrials, Technology, Telecommunications, Utilities, Real Estate and Other.

2. Dataset Characteristics & Scale

2.1. Dataset Classification

- The dataset was collected from 1999 to 2025 and last updated on 28 May 2025.
- Raw dataset: 25.10 GB of .csv files, data collection range: 1999 - 2025
- Working dataset: 12 GB, data collection range: 2000 - 2024
- Processing dataset: 3 GB, data collection: 2000 - 2024

2.2. Dataset format

Column	Type	Description
timestamp	datetime64[ns, UTC]	ISO format datetime (15-minute intervals)
open	float64	Opening price for the interval
high	float64	Highest price during the interval
low	float64	Lowest price during the interval
close	float64	Closing price for the interval
volume	int64	Trading volume for the interval
ticker	string	Stock ticker symbol

3. Data Quality and Preparation

The data quality assessment identified several characteristics of the stock-market dataset. A subset of the dataset, which is proportioned by how the stock industry is represented in the raw, working and processing dataset. Missing values, duplicate observations, closing price validity and volume validity, timestamp validity were evaluated. The analysis also examined the interval between consecutive observations to determine whether the data generally follows the stated 15-minute interval.

4. Exploratory Data Analysis (EDA)

4.1. Data Structure: The working dataset has 3332 files which represent stock prices time intervals of tickers in 12 sectors. The total number of stocks or tickers is 3332. The total data size is 12 GB.

4.2. Sector distribution: It was observed from figure 1 in the appendix that the largest size sector is Financials. Figure 2 shows that 42.4% of the distribution is made of two sectors which are Financials and consumer discretionary, followed by Health Care at 13%, Industrials at 11.7% and the rest make up 32.9%.

4.3. **Data quality:** There are no missing values or duplicates in the dataset. Price (High, Low, Close), volume and timestamp were valid as there were no errors.

4.4. **Temporal Analysis:** Date ranges and time intervals of stocks (tickers) across different sectors are inconsistent. As shown in Table 1 in the appendix, ABAT stock in basic materials date range is between 2023 and 2024 while ADTN in Utilities is between 2000 and 2024. Time intervals range from 15, 60, 120 min to extreme intervals of 705 min.

4.5. **Price Analysis:** The majority of stocks have mean log transformed stock prices that range between 0 and 5 dollars as shown in Figure 3 in the appendix.

4.6. **Volume Analysis:** The log transformed mean trading volume for most stocks or tickers ranges between 7 and 11.5.

4.7. **Correlation Analysis:** A price level characteristic correlation matrix was done as shown in figure 5 in the appendix. There's a strong correlation between mean volume and total volume. There's no correlation between mean volume and mean closing price which indicates that the two metrics move independently.

5. Seven Vs of Big Data

5.1. **Volume** refers to the massive size of the data, and the stock-market data has been generated since 2000-01-03 to 2024-12-03 from more than 6000 different stocks at a 15-minute interval meaning millions of records which are 25.1GB exist. This qualifies the data as having massive volumes for Big Data.

5.2. **Velocity** refers to the speed at which the data is generated and processed. The data is generated at 15-minutes intervals for different stocks. Dashboards would have to be regularly updated qualifying the data as above average velocity.

5.3. **Variety** refers to how the data represents different types of data. The stock market data has timestamps, text, and numerical data types from different stocks which qualifies it for Big Data analysis.

5.4. **Veracity** reflects the reliability, accuracy, and trustworthiness of data. The stock-market data chosen has shown high accuracy and reliability on the data quality checks. The data source is a reputable source and the data behaves in accordance to normal stock-market data on validations.

5.5. **Value** covers the potential insights and benefits derived from the data. Historical stock-market data is useful to predict future movements of stocks. The data is highly useful for investors, market traders and stock market researchers.

5.6. **Variability** captures the inconsistency and fluctuations in data flow or meaning over time. The stock-market data is expected to change over time in respect to the world state. Political and trade stances play a huge role in how the stock-market data changes over time.

5.7. **Visualisation** bridges the gap between Big Data characteristics and human understanding by summarising the large volumes of data. The stock-market data can be visualised through graphs or trend analysis charts. Dedicated candlestick symbols exist to visualise the 4 values of each stock transaction.

6. Value & Limitations

6.1. **Values:** The dataset has historical depth of 20+ years supporting long-term analysis and back-testing. The dataset also has high quality and combines different stocks. This allows for deeper trend analysis and enhanced market insights. Can be used to enhance risk management and regulatory compliance, and faster decision making allowing automated investment strategies and operational efficiency. Investors can also gain competitive advantage and policy makers can use the insights to set policies that would benefit society.

6.2. **Limitations:** The stock data might have a bias towards two sectors (Financials and Consumer discretionary) and that make up almost half of the stock portfolio. The dataset time interval is not consistent with the defined 15-minute (i.e. intervals range from 15 minutes to 1 hour 30 minutes) which might lead to incorrect trend analysis where some data points are not represented.

Appendix

- [Github repository link]

Tables and Figures

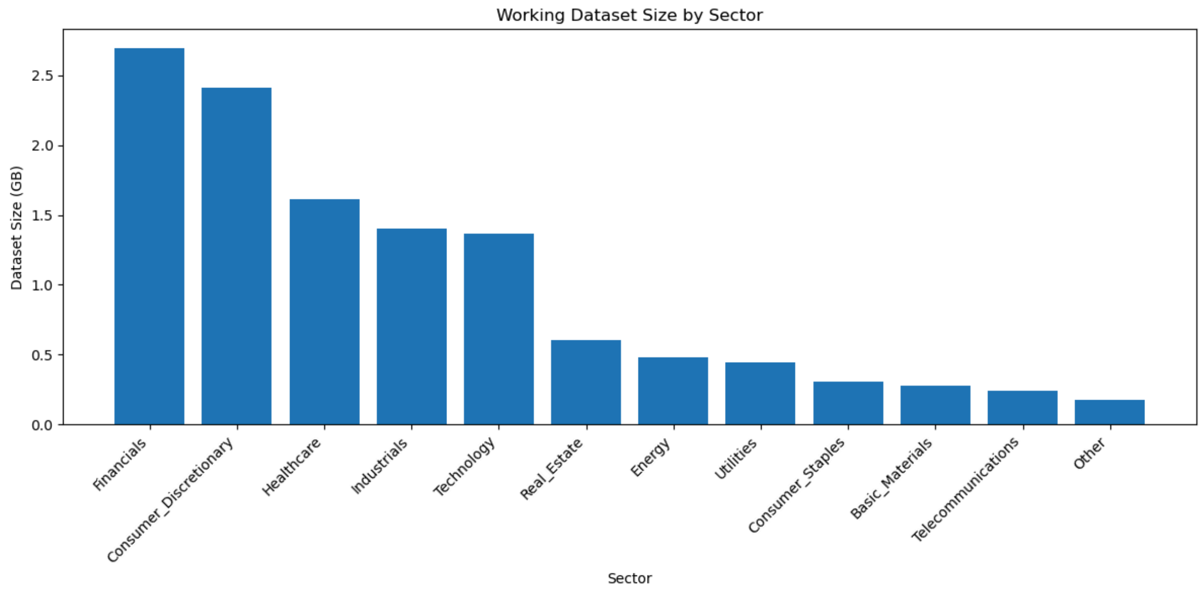

Figure 1: Dataset Size by Sector

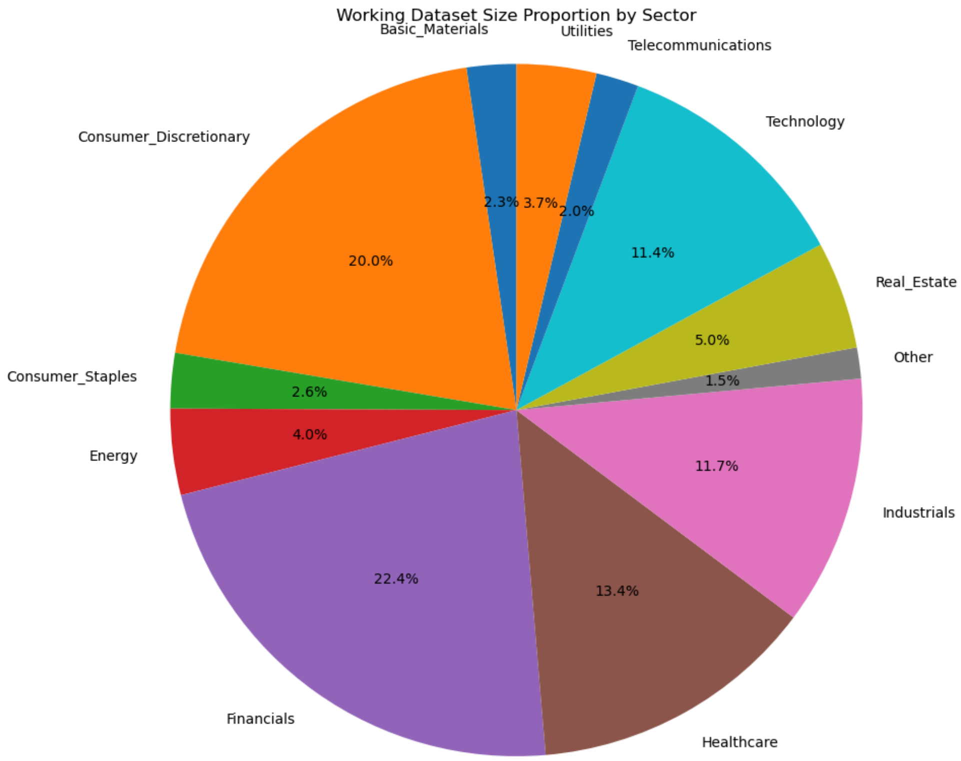

Figure 2: Sector proportion

Table 1: Temporal Analysis of tickers in various sectors

	rows	min_timestamp	max_timestamp	interval_counts	sector	ticker
0	11532	2023-11-02 07:15:00	2024-12-05 19:45:00	{15.0: 9701, 30.0: 757, 60.0: 189, 705.0: 9	Basic_Materials	ABAT
65	135121	2013-12-09 08:30:00	2024-12-03 19:45:00	{15.0: 125728, 30.0: 3525, 660.0: 30, 60.0: 68	Consumer_Discretionary	AAL
681	94579	2013-11-11 09:30:00	2024-12-05 19:45:00	{15.0: 86041, 120.0: 149, 780.0: 28, 60.0: 698	Consumer_Staples	ABEV
757	46810	2017-05-19 15:45:00	2024-12-04 19:00:00	{15.0: 39540, 60.0: 598, 3885.0: 3, 30.0: 2040	Energy	AMPY
850	135283	2006-02-27 09:30:00	2024-12-05 19:30:00	{15.0: 126314, 1065.0: 33, 1050.0: 1990, 393	Financials	AB
1596	49569	2017-08-08 11:45:00	2024-12-03 19:30:00	{2745.0: 1, 1455.0: 2, 45.0: 1109, 5805.0: 1, 4	Healthcare	AADI
2194	183970	2000-01-03 09:30:00	2024-12-03 19:45:00	{15.0: 170986, 75.0: 501, 30.0: 2726, 945.0: 1	Industrials	A
2483	148113	2002-12-16 09:30:00	2024-11-29 13:00:00	{30.0: 3923, 45.0: 1218, 15.0: 135781, 210.0:	Other	ACTG
2673	152737	2004-05-06 09:30:00	2024-11-29 17:00:00	{45.0: 1504, 15.0: 138752, 30.0: 4902, 1065.0	Real_Estate	ABR
2823	90806	2013-09-26 11:15:00	2024-12-03 19:45:00	{15.0: 82491, 45.0: 977, 915.0: 37, 75.0: 339,	Technology	AAOI
3185	3736	2024-07-01 09:00:00	2024-12-03 19:45:00	{15.0: 3243, 120.0: 14, 885.0: 4, 1020.0: 3, 45	Telecommunications	AIOT
3238	181583	2000-01-03 09:30:00	2024-12-03 19:45:00	{15.0: 169691, 1050.0: 1260, 1065.0: 23, 30.0	Utilities	ADTN

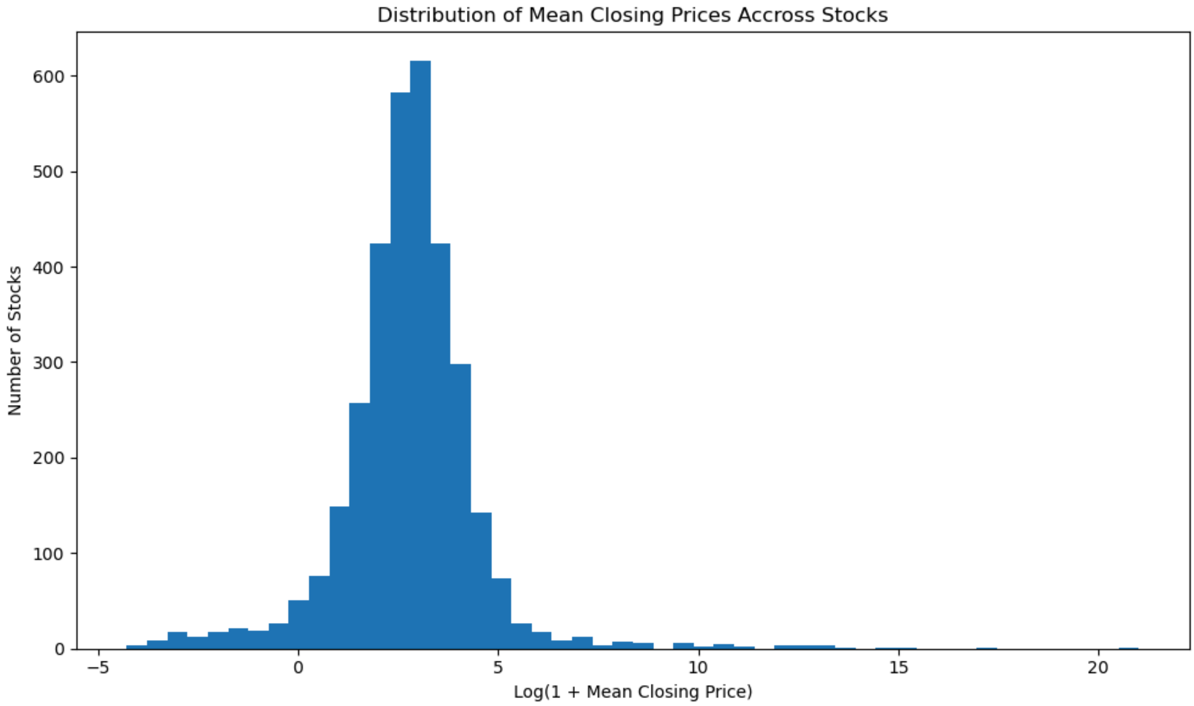

Figure 3: Mean stock prices

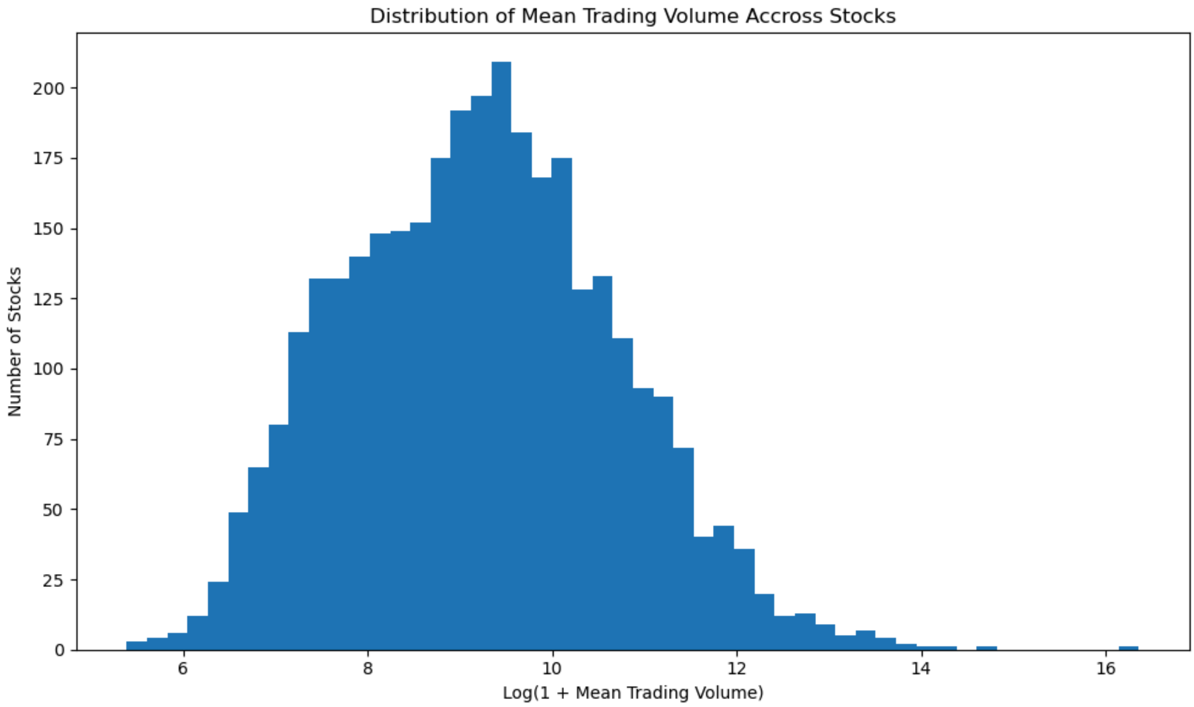

Figure 4: Mean Stock trading volume

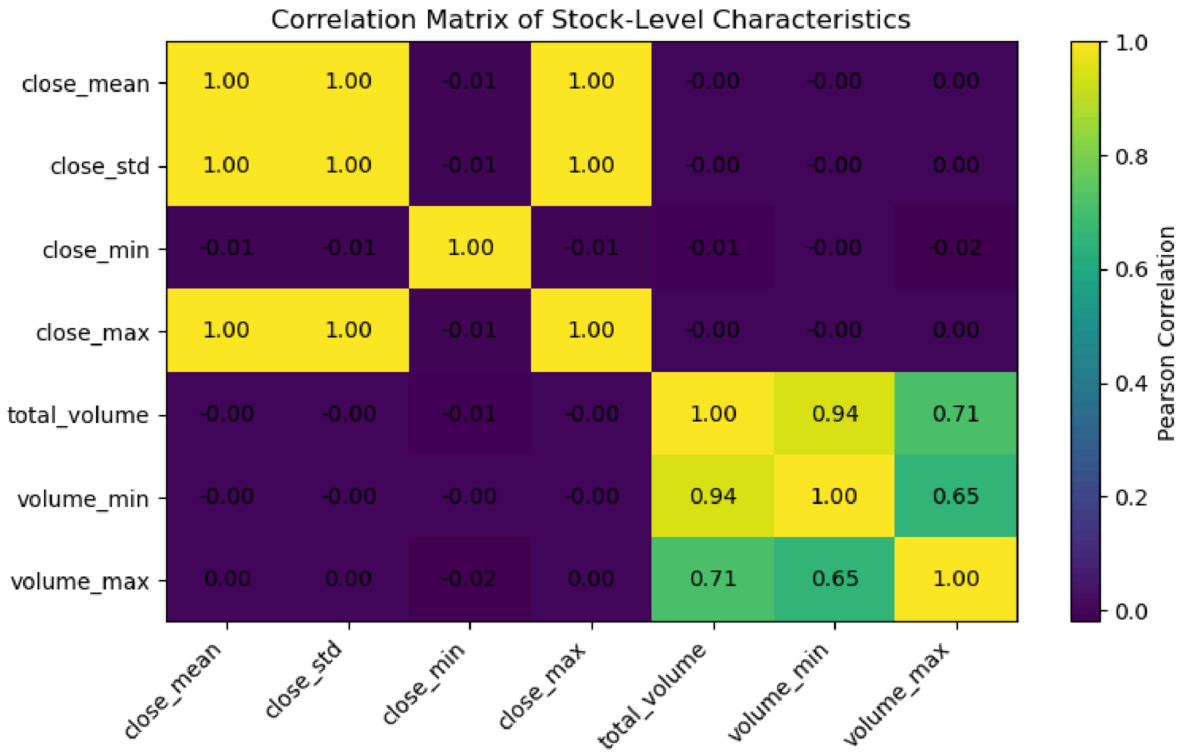

Figure 5: Stock level characteristics correlation matrix